

SMaRT Platform — User Guide

Operations, Maintenance, Warehouse & Resource Tracking System

1. Overview & Core Concepts

SMaRT (System for Operations, Maintenance, and Resource Tracking) is a unified enterprise management application built for high-tempo operations, equipment maintenance, warehouse logistics, and workforce scheduling. Built on an event-driven architecture, every change in one module instantly updates related indicators across the platform.

- Reactive Synchronization:** Updating a work order, sortie, or inventory quantity updates dashboard KPI cards and graphs immediately.
- Audited Workflows:** Every maintenance record, stock transaction, and shift adjustment is logged with actor and timestamp audit trails.
- Role-Based Visibility:** Administrators can fine-tune section and action visibility per user role.

Core Operational Synergy

Operations schedule sorties " Maintenance prepares hardware and logs downtime " Warehouse reserves parts and logs audited stock movements " Shifts ensure qualified technicians are on duty " Analytics synthesizes cross-domain metrics into actionable intelligence.

2. Global Controls & Platform Features

The following core tools are always accessible in the application shell header:

- Global Date Range Filter:** Controls the active time window across Dashboard, Operations, Maintenance, Warehouse, Shifts, and Analytics simultaneously. Offers 1-click presets (Today, 7D, 30D, 90D) and a custom calendar date picker.
- Multi-Format Export Engine:** Download reports on demand with branded headers in Excel (.xlsx), Word (.docx), PDF (.pdf), and PowerPoint (.pptx).
- Real-Time Notification Center:** Live WebSocket alert feed for critical equipment breakdowns, low stock warnings, and overdue task deadlines.
- User Profile & Session Controls:** Manage personal details, avatar, and password via the user menu in the sidebar footer.

3. Section-by-Section Functional Guide

3.1 Dashboard (/dashboard)

The command center displaying live operational health metrics and immediate action items.

Feature Area	Capabilities & User Actions
KPI Metric Cards	Live counters: Active Operations, Pending Maintenance, Low Stock Items, Open Tasks, and Registered Users.
Urgent Action Center	Direct links to Overdue Tasks, Critical Maintenance work orders, and Out-of-Stock inventory items.
Telemetry Charts	Visual trend charts for daily operations volume, maintenance workload distribution, and resource allocations.
Executive Export	Download comprehensive multi-format dashboard summary reports.

3.2 Operations & Sorties (/operations)

Plan, dispatch, and track tactical missions, sorties, and technical deployments.

Feature Area	Capabilities & User Actions
Sortie Lifecycle	Track states: PLANNED "" IN_PROGRESS "" COMPLETED (or ON_HOLD / CANCELLED).
Priority Matrix	Classify urgency with LOW, MEDIUM, HIGH, or CRITICAL badges.
Sortie Details	Specify operation type, team lead, operational location, scheduled dates, and mission debrief notes.
Filters & Export	Filter by date range, priority, and status; export mission logs to Excel, Word, or PDF.

3.3 Maintenance (/maintenance)

Maintains hardware readiness through dual capabilities: active corrective logs and recurring preventive maintenance.

- **Maintenance Records:** Log work under PREVENTIVE, CORRECTIVE, PREDICTIVE, UPGRADE, or INSPECTION. Track hardware asset ID, technician, downtime hours, financial costs, parts used, and technical findings.
- **Planned Maintenance Calendar:** Create recurring routines repeating by DAY, WEEK, or MONTH. Interactive monthly calendar features color-coded indicator dots. Supports master tasks vs. occurrence overrides and checklist generation.

3.4 Warehouse & Inventory (/warehouse)

Asset inventory tracking, catalog management, and audited component movements.

Feature Area	Capabilities & User Actions
Item Catalog	SKU, part number, category, description, unit of measure, unit cost, and supplier information.
Stock Health	Automated warnings when quantities reach Reorder Point or fall below Min Stock thresholds.
5-Type Movements	Log audited physical stock movements: IN (Receipt), OUT (Issue), TRANSFER (Relocate), ADJUSTMENT (Audit count), and RETURN (Unused item).
Audit Ledger	Full transaction history recording quantity delta, remaining stock, user ID, and timestamp.

3.5 Shift Management (/shifts)

Manage workforce rosters, duty rotations, and attendance tracking.

Feature Area	Capabilities & User Actions
Weekly Calendar Grid	Interactive weekly grid displaying scheduled duty assignments across all personnel.
Shift Archetypes	Configurable shift types (Morning, Evening, Night, On-Call) with standard hours and colors.
Time & Attendance	Record clock-in and clock-out timestamps; compute accrued overtime automatically.
Status Tracking	Manage states: SCHEDULED, CONFIRMED, IN_PROGRESS, COMPLETED, ABSENT, or SWAPPED.

3.6 Analytics & Intelligence (/analytics & /analytics/operations)

Business intelligence, operational velocity, and equipment reliability analytics.

- **System Analytics:** Interactive Chart.js visualizations for mission success rates, equipment downtime breakdown, inventory consumption velocity, and shift coverage.
- **Operations Analytics:** Dedicated drill-down dashboard (/analytics/operations) for sortie duration distributions, priority weightings, and location heatmaps.

3.7 Task Manager (/tasks)

Task orchestration supporting both Kanban workflows and deep checklist hierarchies.

- **Dual Views:** Toggle freely between an interactive Kanban board (TODO, IN_PROGRESS, REVIEW, DONE) and a hierarchical List View.
- **Recursive Subtasks:** Nest unlimited subtasks beneath parent deliverables with real-time aggregate completion percentages.

- **Interval Recurrence:** Automate recurring work with intervals: ONCE, DAILY, WEEKLY, MONTHLY, or YEARLY.

3.8 Wiki & Documentation (/wiki)

Centralized engineering knowledge base, standard operating procedures (SOPs), and manuals.

- **Category Tree:** Multi-level hierarchical category tree with dedicated icons for fast document browsing.
- **Markdown Authoring:** Full rich text and markdown editor with code highlighting, checklists, tags, and pinned articles.
- **Article Governance:** Manage document lifecycles (DRAFT " REVIEW " PUBLISHED " ARCHIVED) with version counters.

3.9 Admin Settings (/admin)

Restricted administrative portal for system oversight, security, and master data management.

Admin Tab	Administrative Capabilities
System Overview	Server health, MySQL database connection status, memory consumption, and uptime.
User Directory	Create and modify user profiles, assign roles, toggle active status, and perform secure password resets.
Section Access	Configure sidebar visibility and operational permissions per module for each user role.
Reference Data	Manage Operation Types, Shift Types, Asset Categories, Item Categories, and Warehouse Storage Locations.

4. Roles & Permissions Matrix

SMaRT implements fine-grained Role-Based Access Control (RBAC):

Module / Feature	Admin	Manager	Tech	Viewer
Dashboard	Full Access	Full Access	Full Access	View Only
Operations & Sorties	Full (CRUD)	Full (CRUD)	Update Status	View Only
Maintenance Records	Full (CRUD)	Full (CRUD)	Log & Update	View Only
Planned Maintenance	Full (CRUD)	Full (CRUD)	Mark Done	View Only
Warehouse Inventory	Full (CRUD)	Full (CRUD)	Movements	View Only
Shift Scheduling	Full (CRUD)	Full (CRUD)	Clock In/Out	View Only
Analytics & Reports	Full Access	Full Access	View Only	View Only
Task Manager	Full (CRUD)	Full (CRUD)	Full (CRUD)	View Only
Wiki & Docs	Full (CRUD)	Full (CRUD)	Edit & Create	Read Only
Admin Settings	Full Access	Restricted	No Access	No Access
Multi-Format Export	Enabled	Enabled	Enabled	Enabled

5. Quick Tips for New Users

- **Check Date Filters First:** If a table appears empty, check the date filter in the top header. Click 30D or Clear to reveal all historical records.
- **Log Consumed Spare Parts:** Always attach consumed inventory items when closing maintenance logs to ensure automatic stock decrement.
- **Export Ahead of Briefings:** Use Export " PowerPoint or PDF from Operations or Analytics to instantly create slides and executive summaries.
- **Need Standard Procedures?:** Visit /wiki for official checklists, operating manuals, and technical SOPs.